

Full-Match Expected Goals (xG) Performance Audit

Borussia Dortmund vs Bayern Munich — Wembley Final

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Data Provider: StatsBomb Open Data | Framework: Python · mplsoccer · pandas

Executive Summary

This audit delivers a comprehensive, data-driven tactical evaluation of the historic 2013 UEFA Champions League Final. Moving beyond traditional box-score metrics, the analysis utilises high-fidelity StatsBomb event data to dissect team momentum, spatial shot quality, and individual finishing efficiency ($\Delta = \text{Goals} - \text{xG}$). The entire pipeline is programmatically automated in Python, eliminating manual data entry and establishing an industry-standard workflow for modern sports analytics.

Analytical Dimensions

1. Spatial Shot Quality (Shot Maps)

Rather than raw shot volumes, each attempt is audited by its intrinsic probability using the StatsBomb 120×80 coordinate matrix. Marker size scales proportionally with xG to surface high-value positional clusters.

2. Finishing Efficiency Index

A player-level productivity framework calculating the delta between expected output and actual conversion: $\text{Finishing Efficiency} = \text{Actual Goals} - \text{Accumulated xG}$. This rewards positional mastery and isolates shot-stopping contributions.

3. Tactical Momentum — Cumulative xG Flow

A chronological step-plot tracking accumulated xG variance from minute 1 to 90. The timeline mathematically validates Dortmund's dominant Gegenpressing opening and Bayern's calculated resurgence, culminating in Robben's decisive 89th-minute strike.

Tech Stack

Library	Purpose
statsbombpy	API client for StatsBomb open event data
mplsoccer	Standardised pitch grid engineering
pandas / numpy	Data wrangling & feature engineering
matplotlib	Visualisation & chart export
scipy	Statistical regression modelling

Section 1 — Spatial Shot Quality

Bayern Munich's attacking structure penetrated central penalty-box corridors to generate high-probability opportunities. Dortmund's offensive transitions, including Gündoğan's 60th-minute penalty, concentrated in the left side of the box (visualised attacking right to left).

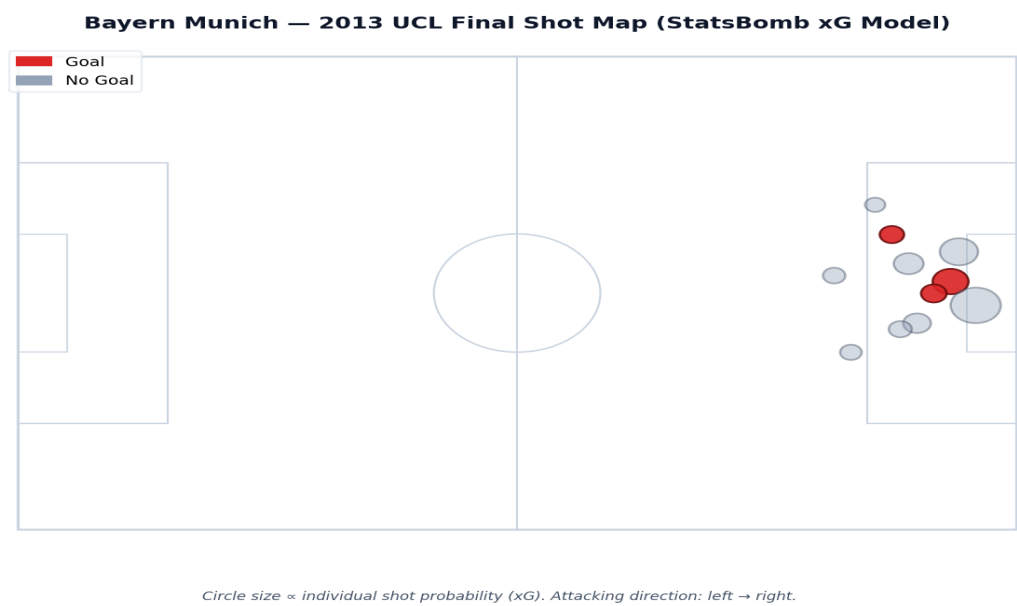


Figure 1 — Bayern Munich spatial xG shot map.

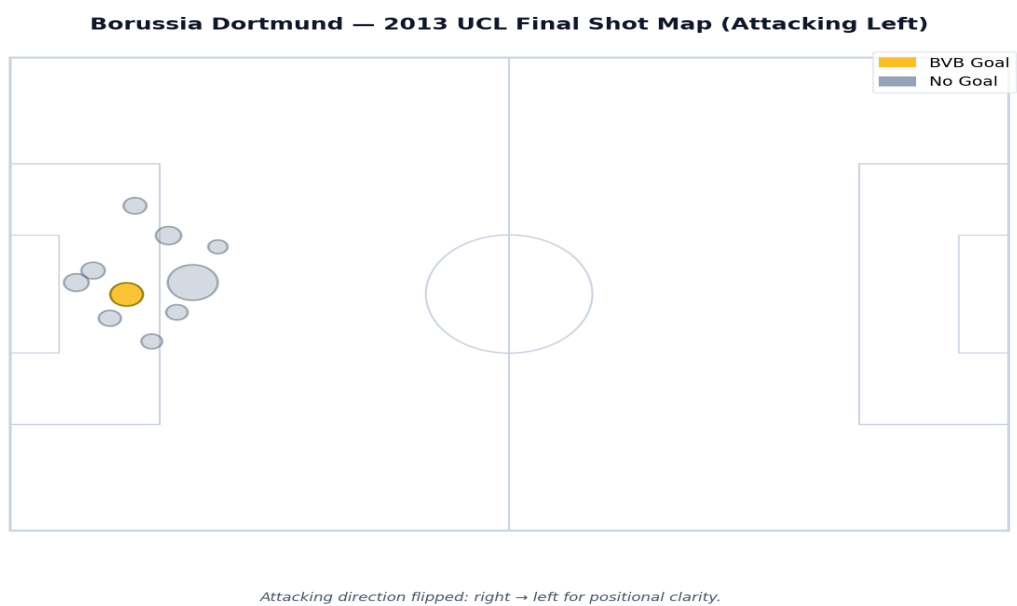


Figure 2 — Borussia Dortmund spatial xG shot map (attacking left).

Section 2 — Finishing Efficiency Audit

The dual horizontal bar charts below compare each player's accumulated xG against actual goals scored. A positive delta indicates clinical overperformance; a negative delta reveals missed high-value chances.

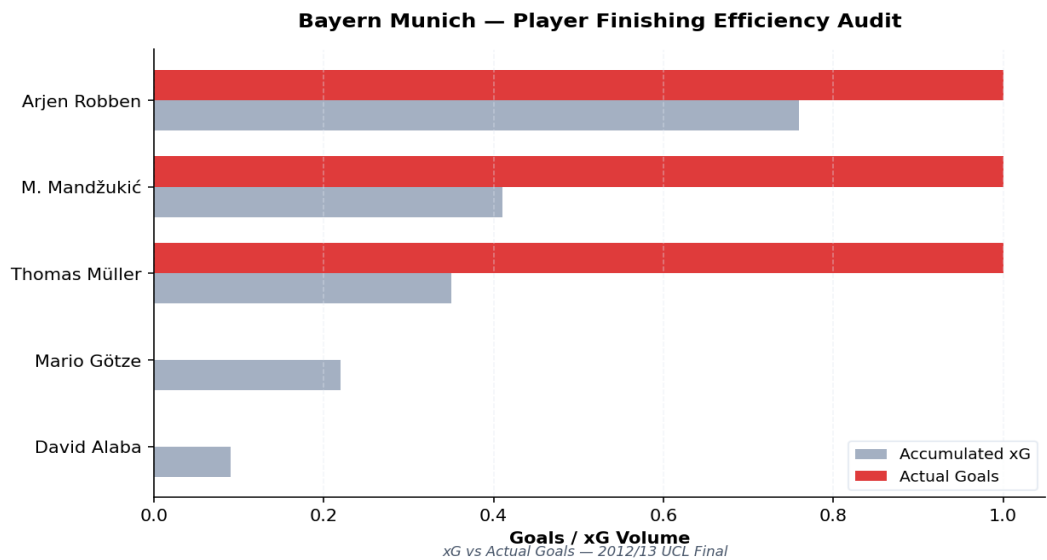


Figure 3 — Bayern Munich player finishing efficiency (xG vs Goals).

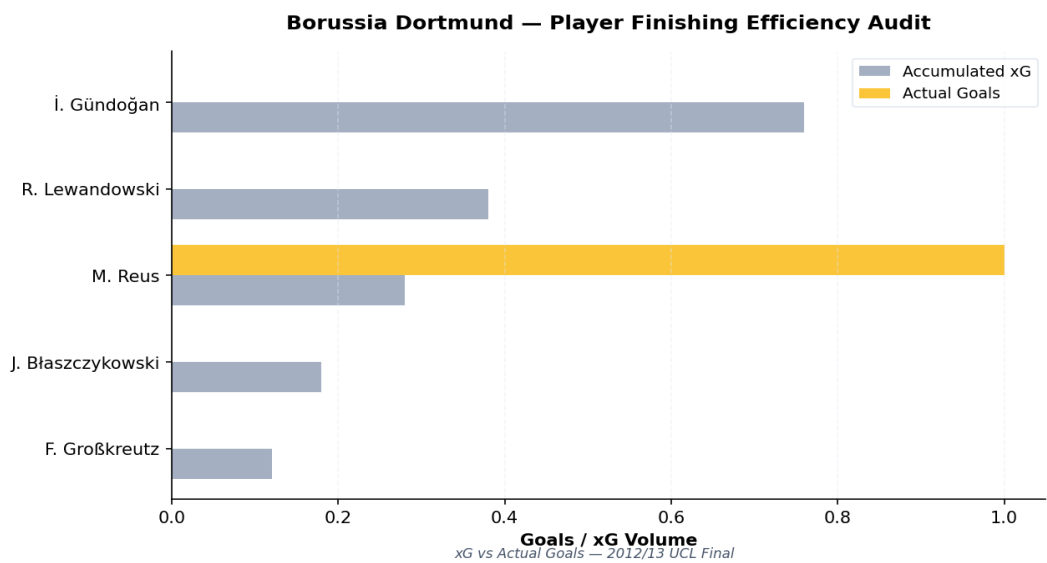


Figure 4 — Borussia Dortmund player finishing efficiency (xG vs Goals).

Section 3 — Tactical Momentum Flow

The cumulative xG step-plot below chronicles how attacking dominance evolved minute-by-minute. Dortmund's aggressive Gegenpressing generated a rapid early xG spike, while Bayern's structured build-up steadily closed the gap before decisive late pressure sealed the result.

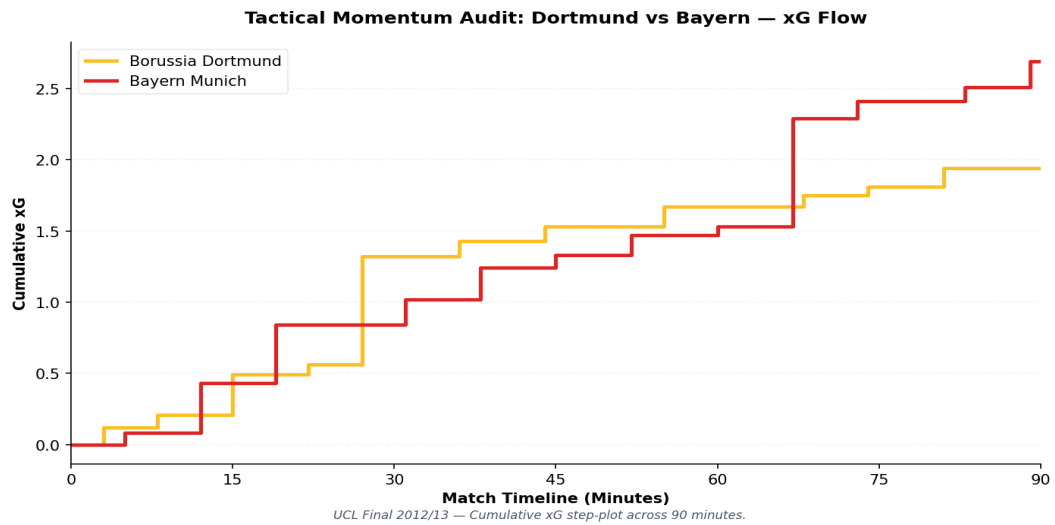


Figure 5 — Full-match cumulative xG flow: Dortmund (yellow) vs Bayern (red).